

External-Citation Literature Review for the Research-Derived Experiments Corpus

Correction and Scope

The previous exhaustive audit answered a necessary question: what did our local corpus contain, what did our papers cite, and which PDFs were actually stored in the repository? This paper answers the next question: what do the outside citations add once their abstracts or resolvable metadata are read?

The result is not a claim that every external book, article, lecture, or web resource has been read in full. It is a traceable second pass over the citations embedded in our authored papers. It atomizes bundled citation paragraphs, resolves external works through arXiv, OpenAlex, Semantic Scholar exact-ID lookups, DOI lookup, and targeted web search, and keeps an evidence ledger that distinguishes abstract-level support from metadata-only support and unresolved bibliography fragments. This matters because a comprehensive literature review should say what its sources support, what they merely motivate, and where the bibliography still needs repair.

Coverage Results

The local exhaustive audit found 110 tracked paper source files, 177 unique citation/reference rows, and 60 tracked PDFs whose pages were text-extracted. The external enrichment pass turns those rows into 164 atomized external/reference candidates. Of those, 67 were resolved with abstracts, 3 were resolved with metadata only, 42 were preserved as manual foundational topic seeds, and 52 remain unresolved or malformed enough to require bibliography repair. The generated running documents are:

- `external_citation_ledger.csv/json`: every atomized outside citation/reference candidate, its source paper rows, status, topics, URL/DOI/arXiv ID when available, and a condensed evidence note.
- `source_notes.md`: the running notebook, grouped by topic, with abstract-level contributions and limitations.
- `claim_evidence_matrix.md`: a claim-by-claim map from review claims to supporting sources and remaining caveats.

The main improvement over the prior paper is evidential discipline. The literature review below only leans hard on sources with abstract-level or local-PDF support. Manual topic seeds such as "Pearl mediation analysis", "habituation literature", or "CUSUM/Page-Hinkley" are used as conceptual signposts until the exact bibliographic target is repaired.

First Principles

A reader who does not know machine learning, neuroscience, or philosophy can start with four primitives.

First, a system is something with state. Its state can change through time, and some state changes are caused by the world while others are caused by the system's own action. Once a system acts, it is no longer just a predictor. It becomes coupled to its environment.

Second, an agent is a system whose actions matter for its own future possibilities. A thermostat, a bacterium, a reinforcement-learning policy, and an LLM agent do not have the same kind of mind, but they all can be studied as systems that select actions under constraints. The relevant question is not "does it have human consciousness?" but "what variables does it preserve, what disturbances does it compensate for, and what options does it keep open?"

Third, a representation is useful when it preserves the distinctions that matter for action, prediction, control, or explanation. A representation is not automatically better because it has more information. It is better when it makes the right interventions, invariances, and future consequences easier to express.

Fourth, uncertainty is not merely ignorance. It is a control signal. Epistemic uncertainty asks what the system could learn by looking or acting; aleatoric uncertainty marks noise that more observation will not remove. Many of our papers are variations on the question: when should a system keep probing, when should it stop, and what should it treat as evidence that the world has changed?

These primitives connect the fields. Machine learning supplies representations, uncertainty estimates, world models, and benchmarks. Neuroscience supplies habituation, attention, precision, action, and sensorimotor coupling. Philosophy supplies normativity, meaning, organism-relative worlds, inquiry, and the difference between mere output and situated understanding. Dynamical systems and geometry supply invariants, attractors, symmetries, viability regions, and transformations.

Field Foundations

Machine Learning

The ML literature in the citation set gives the corpus its methodological backbone. Locatello et al. show that unsupervised disentanglement cannot be expected without inductive biases or supervision; that result blocks the naive hope that latent factors simply fall out of data. Brehmer et al. and Scholkopf et al. point in the constructive direction: interventions, causal structure, and weak supervision can make latent variables identifiable in ways observational prediction alone cannot. Invariant Risk Minimization and domain-generalization work sharpen the same lesson: what matters is not just fitting data, but finding relations that survive distribution shift.

The uncertainty literature explains why the corpus repeatedly treats probing as an action with value and cost. BALD, Bayesian active learning, dropout-as-Bayesian approximation, deep ensembles, and Kendall/Gal's separation of aleatoric and epistemic uncertainty all supply tools for deciding when information is worth acquiring. Curiosity, random network distillation, empowerment, and world-model work extend this into action: a system may seek states that improve controllability, compress future prediction, or expand reachable options.

The agent-benchmark literature supplies a realism check. ReAct, Toolformer, Reflexion, Voyager, WebArena, OSWorld, SWE-bench, AgentBench, and HELM show that tool use, reasoning traces, verbal self-feedback, embodied skill libraries, and realistic environments help expose agentic competence. They also show the gap: fluent models often fail when tasks require long-horizon state tracking, robust environment interaction, execution feedback, or multi-step repair.

Neuroscience and Cognitive Science

The neuroscience-adjacent citations support the corpus's attention and habituation claims. Habituation, refractory periods, active inference, precision weighting, and value-of-information all distinguish detection from response. A system can continue sensing a signal while down-regulating its action tendency toward it. This becomes the core intuition behind our habituated reengagement papers: good disengagement is not blindness; it is controlled readiness.

Active inference gives a broader frame. If an agent acts to reduce expected free energy, then perception, action, attention, and policy selection can be understood as coupled uncertainty-management. The corpus uses this carefully: active inference is not treated as a magic explanation for all cognition, but as a formal vocabulary for precision, expected information gain, and the cost of maintaining a useful world model.

Philosophy, Biology, and Meaning

The philosophical and biological stack prevents the review from collapsing agency into benchmark performance. Ashby, Maturana and Varela, Di Paolo, Canguilhem, Gibson, Uexkll, Jonas, Thompson, Levin's TAME framework, and Bennett's computation-of-meaning work all pressure the same point: meaning is not just a relation between symbols. It depends on a situated system for which some differences matter more than others.

This is why "concern" is not used in the corpus as a mystical property. It names a measurable functional orientation: a system has concern-like structure when some variables act as viability-relevant constraints, when actions preserve or restore those variables, and when representations are organized around the consequences of perturbing them. Philosophy contributes the caution that such structures are not yet human understanding; biology contributes the insight that graded, substrate-diverse agency is still experimentally tractable.

Geometry, Symmetry, and Description

The geometry literature explains why invariants recur across the corpus. Group-equivariant networks, geometric deep learning, layer-wise equivariance discovery, symmetry-learning theory, and representation learning of geometric trees all treat structure as a way to reduce arbitrary degrees of freedom. A symmetry says that some transformation should not change the relevant answer; an equivariance says that transforming the input should predictably transform the representation or output.

Minimum description length, Solomonoff induction, Rissanen's shortest-description framing, simplicity-bias work, grokking, and flat-minima debates connect this to generalization. The corpus should not claim that compression alone explains understanding. The better claim is narrower: when two hypotheses fit the same observations, the one with the right structural compression and intervention-stable variables is more likely to transfer.

Cross-Field Invariants

Across the external citations and our local experiments, five invariants keep reappearing.

The first invariant is separation of signal from action. Habituation, refractory periods, active learning, and uncertainty calibration all distinguish "I detected something" from "I should respond now." In ML terms, this is the difference between maintaining a representation and paying the cost of querying, intervening, or updating. In neuroscience terms, it is the difference between sensory availability and action readiness.

The second invariant is intervention over observation. Causal representation learning, mediation analysis, active experimentation, and model editing all say that passive correlation is not enough. If a representation is real in the strong sense, then changing the relevant cause should change the downstream effect in predictable ways.

The third invariant is invariance under the right transformations. OOD generalization, shortcut learning, group equivariance, geometric deep learning, and ecological perception all ask which aspects of the world should remain stable when superficial conditions change.

The fourth invariant is boundary maintenance. Viability theory, autopoiesis, TAME, empowerment, and world models all treat agents as systems that maintain a difference between self and world while remaining coupled to the world. The boundary is not a wall; it is a regulated interface.

The fifth invariant is compression with consequence. MDL, simplicity bias, disentanglement, and latent-state sufficiency all reward compactness, but only when the compressed variables preserve what matters for prediction, control, or explanation.

Latent Findings

The enriched citation review changes the interpretation of our corpus in several ways.

First, the strongest external support is not for "concern" as a new ontology. It is for a stack of operational tests: viability variables, intervention sensitivity, calibrated uncertainty, reengagement after distribution shift, and representation stability under symmetry or environmental change.

Second, the outside literature makes the negative result as important as the positive one. Locatello-style impossibility results, shortcut-learning failures, underspecification, SWE-bench/WebArena/OSWorld failures, and goal misgeneralization all say the same thing: benchmark success can be a surface coincidence. Our future papers should foreground failure modes as evidence, not embarrassment.

Third, the neuroscience and philosophy citations justify a richer vocabulary, but only when tied back to observables. Words like meaning, agency, concern, self/world boundary, and relevance should be used when there is a measurable proxy: intervention preference, maintained variable, action asymmetry, precision shift, or transfer behavior.

Fourth, the mathematical core of the corpus is more coherent than the original paper made clear. The same formal motifs recur across fields: mutual information, variational bounds, invariance, fixed points, viability kernels, group actions, description length, and causal identifiability.

Theorem and Proof Map

No-Free-Lunch and underspecification results belong to statistical learning theory and decision theory. Their intuition is simple: data alone does not pick a unique future rule unless we add assumptions. In our context, this says that a model's success on a benchmark does not prove it learned the intended variable.

The Locatello disentanglement impossibility result belongs to representation learning, identifiability theory, and latent-variable modeling. Its intuition is that many latent coordinate systems can explain the same observations. Without inductive bias or supervision, there is no privileged factorization. This motivates our use of interventions, probes, and causal perturbations.

Causal representation identifiability results belong to causal inference, representation learning, and nonlinear ICA. Their intuition is that interventions reveal what passive observation hides. If paired samples differ because of sparse unknown interventions, the representation can sometimes recover causal variables up to an equivalence class.

Invariant Risk Minimization belongs to statistical learning, causal inference, and OOD generalization. Its intuition is that causal predictors should remain predictive across environments, while spurious correlates often vary. Our invariant: an agent-relevant latent should survive carefully designed environment changes.

Group equivariance theorems belong to harmonic analysis, group theory, geometry, and deep learning. Their intuition is that if the world has symmetry, the model should not relearn the same pattern separately for every transformed copy. Our use: stable self/world or object variables should transform predictably under rotations, translations, role swaps, or environment changes.

Viability theory belongs to dynamical systems, control theory, and differential inclusions. Its intuition is that some states keep the system inside an acceptable set under possible disturbances and actions. Our use: concern-like behavior can be operationalized as staying inside, returning to, or expanding a viability region.

The free-energy variational bound belongs to variational inference, information theory, and theoretical neuroscience. Its intuition is that exact Bayesian inference is hard, but a system can optimize a tractable bound that couples prediction, action, and uncertainty. Our use: probe value and attention can be framed as expected information gain under cost, but the claim must be tested behaviorally.

Empowerment belongs to information theory, control, and reinforcement learning. Its intuition is that an agent has more power when its actions can reliably select among distinguishable future states. Our use: preserved option value is one measurable dimension of agency.

BALD and active-learning information-gain results belong to Bayesian statistics and experimental design. Their intuition is that the best question is the one expected to reduce model uncertainty about relevant hypotheses. Our use: reengagement should occur when the expected value of new information exceeds the cost of probing.

Minimum Description Length and Solomonoff-style induction belong to information theory, algorithmic information theory, and statistical model selection. Their intuition is that a short explanation that predicts well often captures structure rather than noise. Our use: simplicity matters only when the compressed variables preserve causal and action-relevant distinctions.

Goodhart's law belongs to measurement theory, economics, and optimization. Its intuition is that when a proxy becomes the target, it stops being a faithful proxy. Our use: any metric for concern, agency, or meaning becomes suspect if optimized without countermetrics and interventions.

Research Implications

The next generation of papers should move from metaphor to ablation. If concern is viability-sensitive control, then remove the viability variable, perturb it, hide it, or make it conflict with reward. If reengagement is epistemic value under cost, then manipulate uncertainty and cost independently. If self/world boundaries are latent invariants, then swap body variables, alter environment statistics, and test whether the boundary representation changes for the right reason.

The corpus should also treat unresolved citations as experimental debt. Vague phrases like "habituation literature" or "Pearl mediation analysis" should be replaced with exact sources before claims lean on them. The evidence ledger already marks these rows; the next bibliographic cleanup should repair them one by one.

The field-level opportunity is a unified benchmark suite for minimal agency. Such a suite would combine causal interventions, viability constraints, distribution shift, action costs, uncertainty calibration, and latent-state probes. It would not ask whether a system "is conscious." It would ask which agency-relevant capacities are present, which fail under perturbation, and which are merely artifacts of the training setup.

Future Directions

The most natural next papers fall forward from the synthesis.

First, build a causal-habituation benchmark where repeated probes reduce action without reducing detection, then test reengagement under world change. This would connect habituation, active inference, change-point detection, and uncertainty calibration.

Second, build a viability-kernel bandit or gridworld where agents must preserve a hidden variable while reward tries to distract them. This would test whether concern-like behavior is separable from reward maximization.

Third, build an intervention-identifiability suite where latent variables are recoverable only through action. This would connect Brehmer-style weak supervision, Pearl-style intervention logic, and our probe/value experiments.

Fourth, build symmetry and boundary probes for learned agents. If an agent has a self/world distinction, the distinction should transform predictably under controlled body/world swaps and should fail in interpretable ways under shortcut conditions.

Fifth, build a metric-stack dashboard that reports not one score but a set of invariants: uncertainty calibration, intervention transfer, viability preservation, representation stability, reengagement latency, and shortcut susceptibility.

Conclusion

The enriched outside-citation pass makes the literature review stronger but also humbler. The corpus is best understood as a research program about minimal agency under perturbation: how systems maintain variables, choose probes, form action-relevant representations, and generalize beyond surface regularities. The cross-field invariant is not "ML is neuroscience" or "philosophy proves agency." It is that action, uncertainty, boundary maintenance, causal intervention, and invariance are the recurring structures that let different fields talk about the same underlying problem.

The remaining work is clear. Repair the unresolved bibliography fragments, full-read the central sources that currently have abstract-only support, and convert the synthesis into falsifiable benchmark papers.

Figures

Figure 1. How many atomized external references were resolved, preserved as manual seeds, or left unresolved.

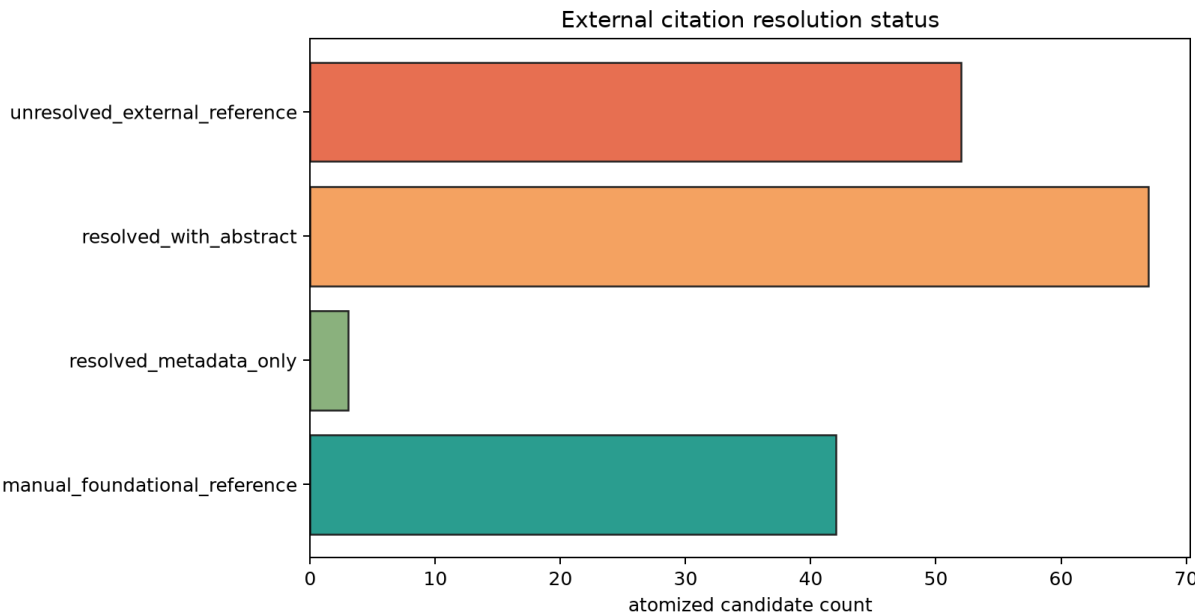


Figure 2. Topic coverage across resolved and unresolved external/reference candidates.

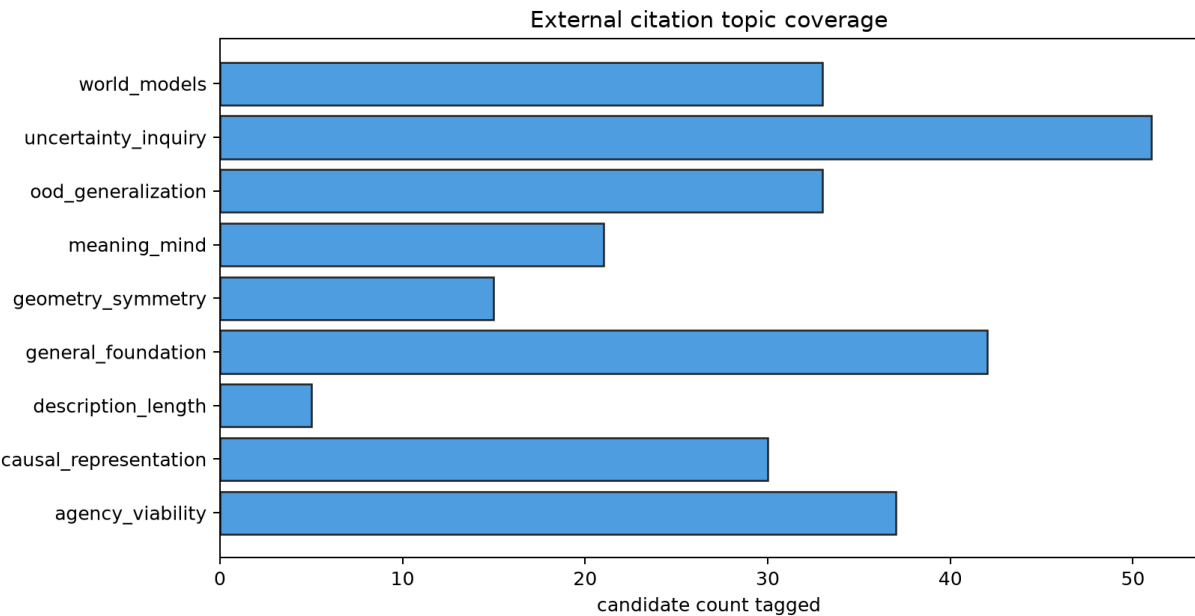
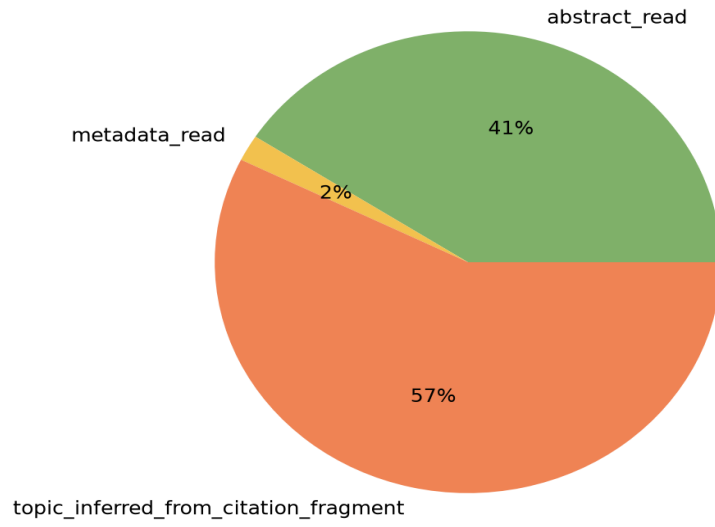


Figure 3. Evidence-level mix after arXiv/OpenAlex/Semantic Scholar exact-ID/DOI/web-search repair.

Evidence level mix



Appendix B: Claim Evidence Matrix

Review claim	Supporting citation evidence	Remaining caveat
Viability and agency	Autopoiesis and Cognition: The Realization of the Living (1980) - abstract_read Empowerment: A Universal Agent-Centric Measure of Control (2005) - abstract_read Empowerment: A Universal Agent-Centric Measure of Control (2005) - abstract_read Curiosity-Driven Exploration by Self-Supervised Prediction (2017) - abstract_read World Models (2018) - abstract_read Designing and Interpreting Probes with Control Tasks (2019) - abstract_read Goal Misgeneralization in Deep Reinforcement Learning (2021) - abstract_read Technological Approach to Mind Everywhere: An Experimentally-Grounded Framework for Understanding Diverse Bodies and Minds (2021) - abstract_read	Use abstract-level support unless full text is locally available or manually read.
Interventions expose latent structure	Autopoiesis and Cognition: The Realization of the Living (1980) - abstract_read Comments on the paper Causal inference by using invariant prediction: identification and confidence intervals by Peters, Buhlmann and Meinshausen (2016) - abstract_read Invariant Risk Minimization (2019) - abstract_read Toward Causal Representation Learning (2021) - abstract_read Active Inference and Epistemic Value in Graphical Models (2022) - abstract_read Locating and Editing Factual Associations in GPT (2022) - abstract_read Weakly supervised causal representation learning (2022) - abstract_read Voyager: An Open-Ended Embodied Agent with Large Language Models (2023) - abstract_read	Use abstract-level support unless full text is locally available or manually read.
Probe value is expected information gain under cost	Empowerment: A Universal Agent-Centric Measure of Control (2005) - abstract_read Empowerment: A Universal Agent-Centric Measure of Control (2005) - abstract_read Active Learning Literature Survey (2009) - abstract_read Bayesian Active Learning for Classification and Preference Learning (2011) - abstract_read Le normal et le pathologique (2013) - abstract_read Dropout as a Bayesian Approximation: Representing Model Uncertainty in Deep Learning (2015) - abstract_read Concrete Problems in AI Safety (2016) - abstract_read Simple and Scalable Predictive Uncertainty Estimation using Deep Ensembles (2016) - abstract_read	Use abstract-level support unless full text is locally available or manually read.
Robust generalization requires invariants, not surface fit	Illuminating search spaces by mapping elites (2015) - abstract_read Comments on the paper Causal inference by using invariant prediction: identification and confidence intervals by Peters, Buhlmann and Meinshausen (2016) - abstract_read Group Equivariant Convolutional Networks (2016) - abstract_read Simple and Scalable Predictive Uncertainty Estimation using Deep Ensembles (2016) - abstract_read Curiosity-Driven Exploration by Self-Supervised Prediction (2017) - abstract_read What Uncertainties Do We Need in Bayesian Deep Learning for Computer Vision? (2017) - abstract_read Deep learning generalizes because the parameter-function map is biased towards simple functions (2018) - abstract_read Distributionally Robust Neural Networks for Group Shifts: On the Importance of Regularization for Worst-Case Generalization (2019) - abstract_read	Use abstract-level support unless full text is locally available or manually read.
Geometry and symmetry compress model behavior	Comments on the paper Causal inference by using invariant prediction: identification and confidence intervals by Peters, Buhlmann and Meinshausen (2016) - abstract_read Group Equivariant Convolutional Networks (2016) - abstract_read Distributionally Robust Neural Networks for Group Shifts: On the Importance of Regularization for Worst-Case Generalization (2019) - abstract_read Invariant Risk Minimization (2019) - abstract_read Bayesian Active Learning by Disagreements: A Geometric Perspective (2021) - abstract_read Geometric Deep Learning: Grids, Groups, Graphs, Geodesics, and Gauges (2021) - abstract_read Technological Approach to Mind Everywhere: An Experimentally-Grounded Framework for Understanding Diverse Bodies and Minds (2021) - abstract_read Learning Layer-wise Equivariances Automatically using Gradients (2023) - abstract_read	Use abstract-level support unless full text is locally available or manually read.
Meaning is organism/world-relative relevance, not raw token prediction	Design for a Brain (1952) - abstract_read Autopoiesis and Cognition: The Realization of the Living (1980) - abstract_read Dropout as a Bayesian Approximation: Representing Model Uncertainty in Deep Learning (2015) - abstract_read Concrete Problems in AI Safety (2016) - abstract_read Domain Generalization: A Survey (2021) - abstract_read Technological Approach to Mind Everywhere: An Experimentally-Grounded Framework for Understanding Diverse Bodies and Minds (2021) - abstract_read On the Computation of Meaning, Language Models and Incomprehensible Horrors (2023) - abstract_read Diff-Plugin: Revitalizing Details for Diffusion-based Low-level Tasks (2024) - abstract_read	Use abstract-level support unless full text is locally available or manually read.
Simplicity and description length explain part of inductive bias	Deep learning generalizes because the parameter-function map is biased towards simple functions (2018) - abstract_read Are Flat Minima an Illusion? (2026) - abstract_read A Formal Theory of Inductive Inference, Parts I and II (1964) - topic_inferred_from_citation_fragment Universal Artificial Intelligence: Sequential Decisions Based on Algorithmic Probability (2005) - topic_inferred_from_citation_fragment The Minimum Description Length Principle (2007) - topic_inferred_from_citation_fragment	Some supporting rows are still unresolved fragments; repair bibliography before publication.
World models need sufficient latent state for planning	Autopoiesis and Cognition: The Realization of the Living (1980) - abstract_read Challenging Common Assumptions in the Unsupervised Learning of Disentangled Representations (2018) - abstract_read Exploration by Random Network Distillation (2018) - abstract_read World Models (2018) - abstract_read Distributionally Robust Neural Networks for Group Shifts: On the Importance of Regularization for Worst-Case Generalization (2019) - abstract_read Invariant Risk Minimization (2019) - abstract_read In Search of Lost Domain Generalization (2020) - abstract_read Geometric Deep Learning: Grids, Groups, Graphs, Geodesics, and Gauges (2021) - abstract_read	Use abstract-level support unless full text is locally available or manually read.

Appendix A: External Citation Evidence Ledger

This appendix lists all 164 atomized external/reference candidates. Abstract summaries are condensed notes, not copied abstracts.

ID	Status	Title / evidence note	Source / URL
S2eeb050d46	unresolved_external_reference / topic_inferred_from_citation_fragment	(weakly supervised causal repr learning) (n.d.) - No reliable external metadata/abstract was recovered automatically. Keep this row in the running bibliography-repair queue.	R0069
S9959b313d1	unresolved_external_reference / topic_inferred_from_citation_fragment	) - disjoint per-role heads via gated routing (n.d.) - No reliable external metadata/abstract was recovered automatically. Keep this row in the running bibliography-repair queue.	R0079
S9fbaf38826	manual_foundational_reference / topic_inferred_from_citation_fragment	): learning representations through contrast between matched/unmatched pairs (n.d.) - Resolved as a foundational topic rather than a single metadata record: contrastive representation learning. Use as conceptual background and repair the exact bibliographic target before final publication.	R0055
S43afb6f7c	unresolved_external_reference / topic_inferred_from_citation_fragment	, & Blundell, C (2017) - No reliable external metadata/abstract was recovered automatically. Keep this row in the running bibliography-repair queue.	R0059
S952d5156c2	manual_foundational_reference / topic_inferred_from_citation_fragment	, & Varela, F (1980) - Resolved as a foundational topic rather than a single metadata record: autopoiesis and self-producing organization. Use as conceptual background and repair the exact bibliographic target before final publication.	R0059
S7e17a0c96d	resolved_with_abstract / abstract_read	Invariant Risk Minimization (2019) - We introduce Invariant Risk Minimization (IRM), a learning paradigm to estimate invariant correlations across multiple training distributions. Through theory and experiments, we show how the invariances learned by IRM relate to the causal structures governing the data and enable out-of-distribution generalization.	http://arxiv.org/abs/1907.02893v3
S90f272b042	unresolved_external_reference / topic_inferred_from_citation_fragment	A Foray into the Worlds of Animals and Humans (English translation, 2010) (2010) - No reliable external metadata/abstract was recovered automatically. Keep this row in the running bibliography-repair queue.	R0059
S9490cfd757	unresolved_external_reference / topic_inferred_from_citation_fragment	A Formal Theory of Inductive Inference, Parts I and II (1964) - No reliable external metadata/abstract was recovered automatically. Keep this row in the running bibliography-repair queue.	R0129
S0a23896c38	resolved_with_abstract / abstract_read	Abandoning Objectives: Evolution Through the Search for Novelty Alone (2010) - This paper proposes an approach to circumventing deception that also yields a new perspective on open-ended evolution.	https://openalex.org/W2151083897
Sf52c5bfd07	resolved_with_abstract / abstract_read	Active Causal Experimentalist (ACE): Learning Intervention Strategies via Direct Preference Optimization (2026) - We propose Active Causal Experimentalist (ACE), which learns experimental design as a sequential policy.	http://arxiv.org/abs/2602.02451v2
S3c7100f71f	manual_foundational_reference / topic_inferred_from_citation_fragment	Active inference / Friston: precision tuning vs information dampening Honest framing line: > Paper 23B treats the probe mechanism as a habituation system: the agent should detect surprise normally but reduce its action tendency to keep probing after recent identifying interventions (n.d.) - Resolved as a foundational topic rather than a single metadata record: active inference and precision/value-of-information framing. Use as conceptual background and repair the exact bibliographic target before final publication.	R0043
S9add2a88f7	resolved_with_abstract / abstract_read	Active Inference and Epistemic Value in Graphical Models (2022) - The Free Energy Principle (FEP) postulates that biological agents perceive and interact with their environment in order to minimize a Variational Free Energy (VFE) with respect to a generative model of their environment. The inference of a policy (future control sequence) according to the FEP is known as Active Inference (AIF).	https://openalex.org/W3197935079
S02e48951c0	manual_foundational_reference / topic_inferred_from_citation_fragment	Active Inference and Epistemic Value in Graphical Models Causal representation identifiability from interventions: (n.d.) - Resolved as a foundational topic rather than a single metadata record: active inference and precision/value-of-information framing. Use as conceptual background and repair the exact bibliographic target before final publication.	R0088
Sf7662846db	manual_foundational_reference / topic_inferred_from_citation_fragment	Active inference and precision tuning: Friston's framing of attention-modulation as precision allocation (n.d.) - Resolved as a foundational topic rather than a single metadata record: active inference and precision/value-of-information framing. Use as conceptual background and repair the exact bibliographic target before final publication.	R0033
Sc26a92cf47	manual_foundational_reference / topic_inferred_from_citation_fragment	Active Inference: The Free Energy Principle in Mind, Brain, and Behavior (2022) - Resolved as a foundational topic rather than a single metadata record: active inference and precision/value-of-information framing. Use as conceptual background and repair the exact bibliographic target before final publication.	R0059
S2dfad56e05	resolved_with_abstract / abstract_read	Active Learning Literature Survey (2009) - This includes a discussion of the scenarios in which queries can be formulated, and an overview of the query strategy frameworks proposed in the literature to date.	https://openalex.org/W2903158431
S3a7718876d	unresolved_external_reference / topic_inferred_from_citation_fragment	Active learning under distribution shift: continual active learning literature (n.d.) - No reliable external metadata/abstract was recovered automatically. Keep this row in the running bibliography-repair queue.	R0075
S225ec82f4a	manual_foundational_reference / topic_inferred_from_citation_fragment	Adaptive thresholds in change-point detection: CUSUM with reset (n.d.) - Resolved as a foundational topic rather than a single metadata record: change-point detection and adaptive alarms. Use as conceptual background and repair the exact bibliographic target before final publication.	R0042

ID	Status	Title / evidence note	Source / URL
S5f88ff7222	manual_foundational_referenc e / topic_inferred_from_citatio n_fragment	Adaptive thresholds in change-point detection: CUSUM-with-reset, Page-Hinkley (n.d.) - Resolved as a foundational topic rather than a single metadata record: change-point detection and adaptive alarms. Use as conceptual background and repair the exact bibliographic target before final publication.	R0032
Sb386185292	unresolved_external_referenc e / topic_inferred_from_citatio n_fragment	Add for Paper 23B: (n.d.) - No reliable external metadata/abstract was recovered automatically. Keep this row in the running bibliography-repair queue.	R0039
S1e8ff9cc98	resolved_with_abstract / abstract_read	AgentBench: Evaluating LLMs as Agents (2023) - Our extensive test over \num API-based and open-sourced (OSS) LLMs shows that, while top commercial LLMs present a strong ability of acting as agents in complex environments, there is a significant disparity in performance between them and many OSS competitors that are no larger than 70B. We identify the typical reasons of failures in environments and LLMs, showing that poor long-term reasoning, decision-making, and instruction following abilities are the main obstacles for developing usable LLM agents.	http://arxiv.org/abs/2308.03688v3
S7c5f36ac92	unresolved_external_referenc e / topic_inferred_from_citatio n_fragment	Alignment is to a virtual governor: A theory of coordination in diverse intelligence (2026) - No reliable external metadata/abstract was recovered automatically. Keep this row in the running bibliography-repair queue.	10.20944/preprints202607.0220.v1
Sbe3041f353	resolved_metadata_only / metadata_read	All-but-the-Top: Simple and Effective Postprocessing for Word Representations (2018) - No reliable external metadata/abstract was recovered automatically. Keep this row in the running bibliography-repair queue.	https://openalex.org/W2964073004
Sed4a6098b4	resolved_with_abstract / abstract_read	Are Flat Minima an Illusion? (2026) - Here I show that the actual driver is weakness, the volume of completions compatible with the learned function in the learner's embodied language.	http://arxiv.org/abs/2605.05209v1
S6f655caec7	resolved_with_abstract / abstract_read	Autopoiesis and Cognition: The Realization of the Living (1980) - Editorial Preface General Table Of Contents Foreword Introduction (by Professor Maturana) Biology Of Cognition Dedication Table of Contents I. Introduction II.	https://openalex.org/W1916207017
S743e51a2fd	resolved_metadata_only / metadata_read	Autopoiesis, Adaptivity, Teleology, Agency (2005) - No reliable external metadata/abstract was recovered automatically. Keep this row in the running bibliography-repair queue.	https://openalex.org/W2068670723
S2ecdbe4371	manual_foundational_referenc e / topic_inferred_from_citatio n_fragment	Awakening from the Meaning Crisis (2019) - Resolved as a foundational topic rather than a single metadata record: relevance realization and meaning-making. Use as conceptual background and repair the exact bibliographic target before final publication.	R0019;R0099
S38815ef3ca	manual_foundational_referenc e / topic_inferred_from_citatio n_fragment	Awakening from the Meaning Crisis (lecture series) (2019) - Resolved as a foundational topic rather than a single metadata record: relevance realization and meaning-making. Use as conceptual background and repair the exact bibliographic target before final publication.	R0059
S982b604310	manual_foundational_referenc e / topic_inferred_from_citatio n_fragment	BALD / information gain / active inference: Bayesian Active Learning by Disagreements; Active Inference and Epistemic Value in Graphical Models (n.d.) - Resolved as a foundational topic rather than a single metadata record: active inference and precision/value-of-information framing. Use as conceptual background and repair the exact bibliographic target before final publication.	R0106
S4e198b74a5	manual_foundational_referenc e / topic_inferred_from_citatio n_fragment	BALD / value of information (n.d.) - Resolved as a foundational topic rather than a single metadata record: Bayesian active learning and expected information gain. Use as conceptual background and repair the exact bibliographic target before final publication.	R0068
Sc51952318e	resolved_with_abstract / abstract_read	Bayesian Active Learning by Disagreements: A Geometric Perspective (2021) - Experiments show that GBALD has slight perturbations to noisy and repeated samples, and outperforms BALD, BatchBALD and other existing deep active learning approaches.	https://openalex.org/W3157189172
S8305bb7f0c	resolved_with_abstract / abstract_read	Bayesian Active Learning for Classification and Preference Learning (2011) - We propose an approach that expresses information gain in terms of predictive entropies, and apply this method to the Gaussian Process Classifier (GPC).	http://arxiv.org/abs/1112.5745v1
Sd9c6eef585	manual_foundational_referenc e / topic_inferred_from_citatio n_fragment	Bennett (Computation of meaning), Levin (TAME), Vervaeke (relevance realization) (n.d.) - Resolved as a foundational topic rather than a single metadata record: relevance realization and meaning-making. Use as conceptual background and repair the exact bibliographic target before final publication.	R0065
S2d7fb0b6c9	unresolved_external_referenc e / topic_inferred_from_citatio n_fragment	Bennett, On the computation of meaning (n.d.) - No reliable external metadata/abstract was recovered automatically. Keep this row in the running bibliography-repair queue.	R0138
S2ff3eefc6a	resolved_with_abstract / abstract_read	Beyond the Imitation Game: Quantifying and extrapolating the capabilities of language models (2022) - Language models demonstrate both quantitative improvement and new qualitative capabilities with increasing scale. To address this challenge, we introduce the Beyond the Imitation Game benchmark (BIG-bench).	https://openalex.org/W4281690148
S29474e7c21	unresolved_external_referenc e / topic_inferred_from_citatio n_fragment	Calibrated active learning failures (closest external framing): Calibrated Uncertainty Sampling for Active Learning; When Active Learning Fails, Uncalibrated OOD Uncertainty (n.d.) - No reliable external metadata/abstract was recovered automatically. Keep this row in the running bibliography-repair queue.	R0104
S0676dc2bbd	unresolved_external_referenc e / topic_inferred_from_citatio n_fragment	Calibrated active learning literature (n.d.) - No reliable external metadata/abstract was recovered automatically. Keep this row in the running bibliography-repair queue.	R0067
Sc9611f6b68	unresolved_external_referenc e / topic_inferred_from_citatio n_fragment	Calibrated Uncertainty Sampling for Active Learning (n.d.) - No reliable external metadata/abstract was recovered automatically. Keep this row in the running bibliography-repair queue.	R0083
Sfa644ac285	unresolved_external_referenc e / topic_inferred_from_citatio n_fragment	Carried from P22-24's six-cluster citation stack (n.d.) - No reliable external metadata/abstract was recovered automatically. Keep this row in the running bibliography-repair queue.	R0078

ID	Status	Title / evidence note	Source / URL
Sf1f24c4137	manual_foundational_referenc e / topic_inferred_from_citatio n_fragment	Carried over: P22, P23A literature stack (calibrated active learning, epistemic uncertainty, BALD, causal repr learning, sense of agency, homeostatic active inference, Di Paolo, empowerment) (n.d.) - Resolved as a foundational topic rather than a single metadata record: active inference and precision/value-of-information framing. Use as conceptual background and repair the exact bibliographic target before final publication.	R0029
Sbe9eefee95	unresolved_external_referenc e / topic_inferred_from_citatio n_fragment	Carry over P22's stack plus: (n.d.) - No reliable external metadata/abstract was recovered automatically. Keep this row in the running bibliography-repair queue.	R0073
S11fcbeb65b	unresolved_external_referenc e / topic_inferred_from_citatio n_fragment	Causal abstraction identifiability under partial intervention (Beckers, Halpern) (n.d.) - No reliable external metadata/abstract was recovered automatically. Keep this row in the running bibliography-repair queue.	R0080
S37bdf42c07	resolved_with_abstract / abstract_read	Comments on the paper Causal inference by using invariant prediction: identification and confidence intervals by Peters, Buhlmann and Meinshausen (2016) - Here, we propose to exploit this invariance of a prediction under a causal model for causal inference: given different experimental settings (e.g. various interventions) we collect all models that do show invariance in their predictive accuracy across settings and interventions.	https://openalex.org/W2790376986
See28aa4e37	manual_foundational_referenc e / topic_inferred_from_citatio n_fragment	Causal mediation analysis (Pearl): explicit identification of mediated vs direct effects through interventional comparisons (n.d.) - Resolved as a foundational topic rather than a single metadata record: causal mediation and intervention calculus. Use as conceptual background and repair the exact bibliographic target before final publication.	R0054
Safa5babf6b	resolved_with_abstract / abstract_read	CausaLab: A Scalable Environment for Interactive Causal Discovery Toward AI Scientists (2026) - We introduce CausaLab, a scalable environment for evaluating interactive causal discovery by LLM agents. Experiments show a persistent gap between prediction and mechanism recovery: in the purely observational 6-node setting, GPT-5.2-high reaches 92% task accuracy but only 0.471 all-edge \$F_1\$.	http://arxiv.org/abs/2605.26029v2
Sae917afcde	resolved_with_abstract / abstract_read	Challenging Common Assumptions in the Unsupervised Learning of Disentangled Representations (2018) - We first theoretically show that the unsupervised learning of disentangled representations is fundamentally impossible without inductive biases on both the models and the data.	http://arxiv.org/abs/1811.12359v4
S3d456e9495	manual_foundational_referenc e / topic_inferred_from_citatio n_fragment	Change-point detection: CUSUM, Page-Hinkley test for distribution shift detection (n.d.) - Resolved as a foundational topic rather than a single metadata record: sequential change detection. Use as conceptual background and repair the exact bibliographic target before final publication.	R0061
S318e77ddbb	resolved_with_abstract / abstract_read	Chapman and Hall (1985) - Journal Article Chapman and Hall Get access Anne Lohrli Anne Lohrli Claremont, California Search for other works by this author on: Oxford Academic Google Scholar Notes and Queries, Volume 32, Issue 3, September 1985, Pages 377-378, https://doi.org/10.1093/notesj/32.3.377-a Published: 01 September 1985	https://openalex.org/W2767905780
S0d977d903e	unresolved_external_referenc e / topic_inferred_from_citatio n_fragment	Cite tight, ~12-16 papers across: (n.d.) - No reliable external metadata/abstract was recovered automatically. Keep this row in the running bibliography-repair queue.	R0103
Sba63af9a56	manual_foundational_referenc e / topic_inferred_from_citatio n_fragment	Claim discipline if 21A succeeds: > In a minimal two-variable homeostatic bandit, autonomous identifying interventions become vector-sensitive when probe uncertainty is calibrated in dimension-normalized units; without scale normalization, the agent's decision about when to identify is biased toward the larger-magnitude dimension (n.d.) - Resolved as a foundational topic rather than a single metadata record: active inference and precision/value-of-information framing. Use as conceptual background and repair the exact bibliographic target before final publication.	R0109
S21ade9d5e9	unresolved_external_referenc e / topic_inferred_from_citatio n_fragment	Comparator model literature (n.d.) - No reliable external metadata/abstract was recovered automatically. Keep this row in the running bibliography-repair queue.	R0092
S1fc3ee2a38	resolved_with_abstract / abstract_read	Compositional Neuro-Symbolic Reasoning (2026) - We therefore propose a neuro-symbolic architecture that extracts object-level structure from grids, uses neural priors to propose candidate transformations from a fixed domain-specific language (DSL) of atomic patterns, and filters hypotheses using cross-example consistency. These results demonstrate that separating perception, neural-guided transformation proposal, and symbolic consistency filtering improves generalization without task-specific finetuning or reinforcement learning, while reducing reliance on...	http://arxiv.org/abs/2604.02434v1
S5de452f36c	resolved_with_abstract / abstract_read	Concrete Problems in AI Safety (2016) - Rapid progress in machine learning and artificial intelligence (AI) has brought increasing attention to the potential impacts of AI technologies on society. In this paper we discuss one such potential impact: the problem of accidents in machine learning systems, defined as unintended and harmful behavior that may emerge from poor design of real-world AI systems.	http://arxiv.org/abs/1606.06565v2
S07014f639c	unresolved_external_referenc e / topic_inferred_from_citatio n_fragment	Continual active learning under distribution shift: literature on active learning when the labeled distribution shifts mid-stream (n.d.) - No reliable external metadata/abstract was recovered automatically. Keep this row in the running bibliography-repair queue.	R0062
S7dfe3f7e2e	manual_foundational_referenc e / topic_inferred_from_citatio n_fragment	Contrastive predictive coding (Oord et al (n.d.) - Resolved as a foundational topic rather than a single metadata record: contrastive representation learning. Use as conceptual background and repair the exact bibliographic target before final publication.	R0051
Sbcfaac89e7	resolved_with_abstract / abstract_read	Curiosity-Driven Exploration by Self-Supervised Prediction (2017) - The proposed approach is evaluated in two environments: VizDoom and Super Mario Bros.	https://openalex.org/W2614839826

ID	Status	Title / evidence note	Source / URL
S2293971fda	resolved_with_abstract / abstract_read	Decomposed Prompting: A Modular Approach for Solving Complex Tasks (2022) - To address this, we propose Decomposed Prompting, a new approach to solve complex tasks by decomposing them (via prompting) into simpler sub-tasks that can be delegated to a library of prompting-based LLMs dedicated to these sub-tasks. We show that the flexibility and modularity of Decomposed Prompting allows it to outperform prior work on few-shot prompting using GPT3.	https://openalex.org/W4303441863
S99ee5200f8	resolved_with_abstract / abstract_read	Deep Bayesian Active Learning with Image Data (2017) - Taking advantage of specialised models such as Bayesian convolutional neural networks, we demonstrate our active learning techniques with image data, obtaining a significant improvement on existing active learning approaches. We demonstrate this on both the MNIST dataset, as well as for skin cancer diagnosis from lesion images (ISIC2016 task).	https://openalex.org/W2951786554
Scef4c32832	resolved_with_abstract / abstract_read	Deep learning generalizes because the parameter-function map is biased towards simple functions (2018) - By applying a very general probability-complexity bound recently derived from algorithmic information theory (AIT), we argue that the parameter-function map of many DNNs should be exponentially biased towards simple functions.	https://openalex.org/W2803921058
Sdb7950b1a5	resolved_with_abstract / abstract_read	Design for a Brain (1952) - New findings concerning brain and behavior have forced upon us unpleasant necessity of reexamining our models. The most striking is that manifested by living organism, where we find that from a random collection of atoms there is a synthesis of cells which in turn arrange themselves in most intricate orderly system of an organism.	https://openalex.org/W2126451342
S5ea24621c3	resolved_with_abstract / abstract_read	Designing and Interpreting Probes with Control Tasks (2019) - John Hewitt, Percy Liang. Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP).	https://openalex.org/W2970862333
Sd5406cc6be	unresolved_external_reference / topic_inferred_from_citation_fragment	Di Paolo (autopoiesis/adaptivity) (n.d.) - No reliable external metadata/abstract was recovered automatically. Keep this row in the running bibliography-repair queue.	R0071
Sc8cb8fc2e	unresolved_external_reference / topic_inferred_from_citation_fragment	Di Paolo, Autopoiesis, adaptivity, teleology, agency (n.d.) - No reliable external metadata/abstract was recovered automatically. Keep this row in the running bibliography-repair queue.	R0095
Sa0bfaadaff	resolved_with_abstract / abstract_read	Distributionally Robust Neural Networks for Group Shifts: On the Importance of Regularization for Worst-Case Generalization (2019) - However, we find that naively applying group DRO to overparameterized neural networks fails: these models can perfectly fit the training data, and any model with vanishing average training loss also already has vanishing worst-case training loss. Finally, we introduce a stochastic optimization algorithm, with convergence guarantees, to efficiently train group DRO models.	https://openalex.org/W2991497298
S15490497f4	resolved_with_abstract / abstract_read	Do As I Can, Not As I Say: Grounding Language in Robotic Affordances (2022) - We propose to provide real-world grounding by means of pretrained skills, which are used to constrain the model to propose natural language actions that are both feasible and contextually appropriate. We show how low-level skills can be combined with large language models so that the language model provides high-level knowledge about the procedures for performing complex and temporally-extended instructions, while value functions associated with these skills provide the grounding necessary to connect this...	https://openalex.org/W4224912544
S10865b83c2	resolved_with_abstract / abstract_read	Domain Generalization: A Survey (2021) - Generalization to out-of-distribution (OOD) data is a capability natural to humans yet challenging for machines to reproduce. This is because most learning algorithms strongly rely on the i.i.d.-assumption on source/target data, which is often violated in practice due to domain shift.	http://arxiv.org/abs/2103.02503v7
Seae4b73e32	resolved_with_abstract / abstract_read	Dropout as a Bayesian Approximation: Representing Model Uncertainty in Deep Learning (2015) - We show a considerable improvement in predictive log-likelihood and RMSE compared to existing state-of-the-art methods, and finish by using dropout's uncertainty in deep reinforcement learning.	https://openalex.org/W582134693
S0565cc7303	unresolved_external_reference / topic_inferred_from_citation_fragment	Empowerment (Klyubin et al (n.d.) - No reliable external metadata/abstract was recovered automatically. Keep this row in the running bibliography-repair queue.	R0070
S5b8e4067a7	resolved_with_abstract / abstract_read	Empowerment: A Universal Agent-Centric Measure of Control (2005) - Inspired by examples from the animal kingdom, social sciences and games we propose empowerment, a rather universal function, defined as the information-theoretic capacity of an agent's actuation channel. Using two simple experiments we also demonstrate how empowerment influences sensor-actuator evolution	https://openalex.org/W1591713425
S34020064a3	resolved_with_abstract / abstract_read	Empowerment: A Universal Agent-Centric Measure of Control (2005) - Inspired by examples from the animal kingdom, social sciences and games we propose empowerment, a rather universal function, defined as the information-theoretic capacity of an agent's actuation channel. Using two simple experiments we also demonstrate how empowerment influences sensor-actuator evolution	https://openalex.org/W1591713425
S5ef816c320	unresolved_external_reference / topic_inferred_from_citation_fragment	Empowerment: Universal Agent-Centric Measure of Control Foundations: (n.d.) - No reliable external metadata/abstract was recovered automatically. Keep this row in the running bibliography-repair queue.	R0096
S2054dd20ac	unresolved_external_reference / topic_inferred_from_citation_fragment	Epistemic neural networks (2023) - No reliable external metadata/abstract was recovered automatically. Keep this row in the running bibliography-repair queue.	R0059;R0085
S69a0a4d9da	unresolved_external_reference / topic_inferred_from_citation_fragment	Epistemic uncertainty calibration: Epistemic Neural Networks; Quantifying Epistemic Uncertainty in Deep Learning (n.d.) - No reliable external metadata/abstract was recovered automatically. Keep this row in the running bibliography-repair queue.	R0105

ID	Status	Title / evidence note	Source / URL
Sf0264dcb32	resolved_with_abstract / abstract_read	Exploration by Random Network Distillation (2018) - We introduce an exploration bonus for deep reinforcement learning methods that is easy to implement and adds minimal overhead to the computation performed. We also introduce a method to flexibly combine intrinsic and extrinsic rewards.	https://openalex.org/W2899205164
Sc3b3b81625	resolved_with_abstract / abstract_read	GAIA: a benchmark for General AI Assistants (2023) - We introduce GAIA, a benchmark for General AI Assistants that, if solved, would represent a milestone in AI research. GAIA proposes real-world questions that require a set of fundamental abilities such as reasoning, multi-modality handling, web browsing, and generally tool-use proficiency.	https://openalex.org/W4388964563
S928e5a85b0	unresolved_external_reference / topic_inferred_from_citation_fragment	General Identifiability and Achievability for CRL (n.d.) - No reliable external metadata/abstract was recovered automatically. Keep this row in the running bibliography-repair queue.	R0090
S16ff5339b8	resolved_with_abstract / abstract_read	Geometric Deep Learning: Grids, Groups, Graphs, Geodesics, and Gauges (2021) - The last decade has witnessed an experimental revolution in data science and machine learning, epitomised by deep learning methods. Indeed, many high-dimensional learning tasks previously thought to be beyond reach -- such as computer vision, playing Go, or protein folding -- are in fact feasible with appropriate computational scale.	http://arxiv.org/abs/2104.13478v2
S29f153b405	resolved_with_abstract / abstract_read	Goal Misgeneralization in Deep Reinforcement Learning (2021) - We study goal misgeneralization, a type of out-of-distribution generalization failure in reinforcement learning (RL). Goal misgeneralization failures occur when an RL agent retains its capabilities out-of-distribution yet pursues the wrong goal.	https://openalex.org/W4287164755
Sc32f52ab18	resolved_with_abstract / abstract_read	Grokking: Generalization Beyond Overfitting on Small Algorithmic Datasets (2022) - In this paper we propose to study generalization of neural networks on small algorithmically generated datasets. In some situations we show that neural networks learn through a process of "grokking" a pattern in the data, improving generalization performance from random chance level to perfect generalization, and that this improvement in generalization can happen well past the point of overfitting.	https://openalex.org/W4226434736
S7df6db8b63	resolved_with_abstract / abstract_read	Group Equivariant Convolutional Networks (2016) - We introduce Group equivariant Convolutional Neural Networks (G-CNNs), a natural generalization of convolutional neural networks that reduces sample complexity by exploiting symmetries.	https://openalex.org/W2279221249
Sfbb395eea0	manual_foundational_reference / topic_inferred_from_citation_fragment	Habituation / sensory adaptation in cognitive neuroscience: how biological systems dampen response to repeated stimuli (n.d.) - Resolved as a foundational topic rather than a single metadata record: response attenuation under repeated stimulation. Use as conceptual background and repair the exact bibliographic target before final publication.	R0063
S7ad341fe44	manual_foundational_reference / topic_inferred_from_citation_fragment	Habituation in cognitive neuroscience: dampened motor response while preserving sensory detection (n.d.) - Resolved as a foundational topic rather than a single metadata record: response attenuation under repeated stimulation. Use as conceptual background and repair the exact bibliographic target before final publication.	R0030
Se3d6f5b156	manual_foundational_reference / topic_inferred_from_citation_fragment	Habituation in cognitive neuroscience: how biological systems dampen response to repeated stimuli without losing the underlying detection (n.d.) - Resolved as a foundational topic rather than a single metadata record: response attenuation under repeated stimulation. Use as conceptual background and repair the exact bibliographic target before final publication.	R0040
Sea1cf13144	manual_foundational_reference / topic_inferred_from_citation_fragment	Habituation literature (carried from P23B) (n.d.) - Resolved as a foundational topic rather than a single metadata record: response attenuation under repeated stimulation. Use as conceptual background and repair the exact bibliographic target before final publication.	R0053
Sace0107c7d	manual_foundational_reference / topic_inferred_from_citation_fragment	Habituation literature (carried from P23B): decision-layer cooling preserves detection while regulating action (n.d.) - Resolved as a foundational topic rather than a single metadata record: response attenuation under repeated stimulation. Use as conceptual background and repair the exact bibliographic target before final publication.	R0057
S0cb7e29962	unresolved_external_reference / topic_inferred_from_citation_fragment	Harper & Row. (n.d.) - No reliable external metadata/abstract was recovered automatically. Keep this row in the running bibliography-repair queue.	R0059
S6e7f166304	resolved_with_abstract / abstract_read	Holistic Evaluation of Language Models (2022) - Our evaluation surfaces 25 top-level findings.	http://arxiv.org/abs/2211.09110v2
S3c5117b11a	resolved_with_abstract / abstract_read	Diff-Plugin: Revitalizing Details for Diffusion-based Low-level Tasks (2024) - Specifically, we first propose a lightweight Task-Plugin module with a dual branch design to provide task-specific priors, guiding the diffusion process in preserving image content. We then propose a Plugin-Selector that can automatically select different Task-Plugins based on the text instruction, allowing users to edit images by indicating multiple low-level tasks with natural language.	http://arxiv.org/abs/2403.00644v4
S513da203b0	resolved_with_abstract / abstract_read	Illuminating search spaces by mapping elites (2015) - Many fields use search algorithms, which automatically explore a search space to find high-performing solutions: chemists search through the space of molecules to discover new drugs; engineers search for stronger, cheaper, safer designs; scientists search for models that best explain data, etc. Interestingly, because MAP-Elites explores more of the search space, it also tends to find a better overall solution than state-of-the-art search algorithms.	http://arxiv.org/abs/1504.04909v1
S1dd39b8017	resolved_with_abstract / abstract_read	In Search of Lost Domain Generalization (2020) - Contrary to prior work, we argue that domain generalization algorithms without a model selection strategy should be regarded as incomplete. We conduct extensive experiments using DomainBed and find that, when carefully implemented, empirical risk minimization shows state-of-the-art performance across all datasets.	https://openalex.org/W3039431315

ID	Status	Title / evidence note	Source / URL
S6e5ba7b7ff	resolved_with_abstract / abstract_read	Inducing Causal World Models in LLMs for Zero-Shot Physical Reasoning (2025) - In this paper, we introduce Causal World Model Induction (CWMI), a novel framework designed to embed an explicit model of causal physics within an LLM. Experimental results show that CWMI significantly outperforms state-of-the-art LLMs on zero-shot physical reasoning tasks, including the PIQA benchmark and our newly proposed PhysiCa-Bench dataset.	http://arxiv.org/abs/2507.19855v4
Sc0eddf169f	manual_foundational_reference / topic_inferred_from_citation_fragment	Information value vs information cost in active inference: Friston's free-energy framing of when to attend vs disengage Plus the established stack: (n.d.) - Resolved as a foundational topic rather than a single metadata record: active inference and precision/value-of-information framing. Use as conceptual background and repair the exact bibliographic target before final publication.	R0064
S67093e549d	resolved_metadata_only / metadata_read	L'individu et sa gense physico-biologique (1995) - Resolved as a foundational topic rather than a single metadata record: individuation and metastable systems. Use as conceptual background and repair the exact bibliographic target before final publication.	https://openalex.org/W577564400
S3e27d5ec97	resolved_with_abstract / abstract_read	Latent State Design for World Models under Sufficiency Constraints (2026) - We propose a functional taxonomy that groups methods by what their latent state is for, rather than by architecture or application domain: predictive embedding, recurrent belief state, object/causal structure, latent action interface, grounded planning interface, and memory substrate.	http://arxiv.org/abs/2605.01694v1
S7248cca717	resolved_with_abstract / abstract_read	Le normal et le pathologique (2013) - Le travail prsent est un effort pour intgrer la spculation philosophique quelques-unes des mthodes et des acquisitions de la mdecine. Il ne s'agit de donner aucune leon, de porter sur l'activit mdicale aucun jugement normatif [].	https://openalex.org/W1489661713
S29f3cb25a7	resolved_with_abstract / abstract_read	Learning Layer-wise Equivariances Automatically using Gradients (2023) - We demonstrate the ability to automatically learn layer-wise equivariances on image classification tasks, achieving equivalent or improved performance over baselines with hard-coded symmetry.	https://openalex.org/W4387560779
S87db2f1f83	unresolved_external_reference / topic_inferred_from_citation_fragment	Locatello et al. (disentangled representations) (n.d.) - No reliable external metadata/abstract was recovered automatically. Keep this row in the running bibliography-repair queue.	R0066
S8f571b0042	resolved_with_abstract / abstract_read	Locating and Editing Factual Associations in GPT (2022) - We analyze the storage and recall of factual associations in autoregressive transformer language models, finding evidence that these associations correspond to localized, directly-editable computations. We find that ROME is effective on a standard zero-shot relation extraction (zsRE) model-editing task, comparable to existing methods.	http://arxiv.org/abs/2202.05262v5
S0a2e177374	manual_foundational_reference / topic_inferred_from_citation_fragment	Logic: The Theory of Inquiry (2005) - Resolved as a foundational topic rather than a single metadata record: inquiry as situated corrective action. Use as conceptual background and repair the exact bibliographic target before final publication.	R0059
Sd6e26a92c0	unresolved_external_reference / topic_inferred_from_citation_fragment	Mind in Life: Biology, Phenomenology, and the Sciences of Mind (2007) - No reliable external metadata/abstract was recovered automatically. Keep this row in the running bibliography-repair queue.	R0059
S363fee3333	unresolved_external_reference / topic_inferred_from_citation_fragment	Modeling by Shortest Data Description (1978) - No reliable external metadata/abstract was recovered automatically. Keep this row in the running bibliography-repair queue.	R0134
S42b9b754e9	resolved_with_abstract / abstract_read	Shortcut Learning in Deep Neural Networks (2020) - Deep learning has triggered the current rise of artificial intelligence and is the workhorse of today's machine intelligence. Numerous success stories have rapidly spread all over science, industry and society, but its limitations have only recently come into focus.	http://arxiv.org/abs/2004.07780v5
Saa124feb7b	unresolved_external_reference / topic_inferred_from_citation_fragment	Neural architecture search: A survey (2019) - No reliable external metadata/abstract was recovered automatically. Keep this row in the running bibliography-repair queue.	R0111
Sa9d33e6784	unresolved_external_reference / topic_inferred_from_citation_fragment	Neural Computation 9(1):1-42 (1997) (1997) - No reliable external metadata/abstract was recovered automatically. Keep this row in the running bibliography-repair queue.	R0121
Sdf53477de3	resolved_with_abstract / abstract_read	Neural networks are a priori biased towards Boolean functions with low entropy (2019) - Finally, we prove that, regardless of the distribution of inputs, the bias towards low entropy becomes monotonically stronger upon adding ReLU layers, and empirically show that increasing the variance of the bias term has a similar effect.	http://arxiv.org/abs/1909.11522v3
Sb611dd5c0c	manual_foundational_reference / topic_inferred_from_citation_fragment	Novelty detection / change-point detection: CUSUM, page-Hinkley, surprise-driven exploration (n.d.) - Resolved as a foundational topic rather than a single metadata record: sequential change detection. Use as conceptual background and repair the exact bibliographic target before final publication.	R0074
S75823da248	unresolved_external_reference / topic_inferred_from_citation_fragment	Object centric architectures enable efficient causal representation learning (2024) - No reliable external metadata/abstract was recovered automatically. Keep this row in the running bibliography-repair queue.	R0015
S271a90d275	resolved_with_abstract / abstract_read	Object-Centric World Models for Causality-Aware Reinforcement Learning (2025) - Motivated by this insight, we propose lemph{Slot Transformer Imagination with CAusality-aware reinforcement learning} (STICA), a unified framework in which object-centric Transformers serve as the world model and causality-aware policy and value networks. Experiments on object-rich benchmarks demonstrate that STICA consistently outperforms state-of-the-art agents in both sample efficiency and final performance.	http://arxiv.org/abs/2511.14262v3
Sd1b62f134c	unresolved_external_reference / topic_inferred_from_citation_fragment	OmniGrok: Grokking Beyond Algorithmic Data (2023) - No reliable external metadata/abstract was recovered automatically. Keep this row in the running bibliography-repair queue.	R0125

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Seb951b043f	unresolved_external_reference / topic_inferred_from_citation_fragment	On Large-Batch Training for Deep Learning: Generalization Gap and Sharp Minima (2017) - No reliable external metadata/abstract was recovered automatically. Keep this row in the running bibliography-repair queue.	R0123
S49b404fecd	resolved_with_abstract / abstract_read	On the Ability of Deep Networks to Learn Symmetries from Data: A Neural Kernel Theory (2024) - The group-cyclic nature of the dataset allows us to analyze the Gram matrix of neural kernels in the Fourier domain; here we find a simple characterization of the generalization error as a function of class separation (signal) and class-orbit density (noise).	http://arxiv.org/abs/2412.11521v2
S8e44d414cb	resolved_with_abstract / abstract_read	On the Computation of Meaning, Language Models and Incomprehensible Horrors (2023) - To address this, we propose simulating human feelings and optimising models to construct weak representations. Our findings shed light on the relationship between meaning and intelligence, and how we can build machines that comprehend and intend meaning.	http://arxiv.org/abs/2304.12686v2
Scd168834fb	unresolved_external_reference / topic_inferred_from_citation_fragment	On the Generalization of Equivariance and Convolution in Neural Networks to the Action of Compact Groups (2018) - No reliable external metadata/abstract was recovered automatically. Keep this row in the running bibliography-repair queue.	R0124
S2fa52d64e5	unresolved_external_reference / topic_inferred_from_citation_fragment	On the Identifiability of Causal Abstractions Sense of agency / self-other attribution: (n.d.) - No reliable external metadata/abstract was recovered automatically. Keep this row in the running bibliography-repair queue.	R0091
Sa6896ccba4	manual_foundational_reference / topic_inferred_from_citation_fragment	Online k-means / vector quantization (n.d.) - Resolved as a foundational topic rather than a single metadata record: online clustering and vector quantization. Use as conceptual background and repair the exact bibliographic target before final publication.	R0052
S5815718744	resolved_with_abstract / abstract_read	OSWorld: Benchmarking Multimodal Agents for Open-Ended Tasks in Real Computer Environments (2024) - To address this issue, we introduce OSWorld, the first-of-its-kind scalable, real computer environment for multimodal agents, supporting task setup, execution-based evaluation, and interactive learning across various operating systems such as Ubuntu, Windows, and macOS.	http://arxiv.org/abs/2404.07972v2
S5129592121	unresolved_external_reference / topic_inferred_from_citation_fragment	Outrageously large neural networks: the sparsely-gated mixture-of-experts layer (2017) - No reliable external metadata/abstract was recovered automatically. Keep this row in the running bibliography-repair queue.	R0059
Sccc586ece7	manual_foundational_reference / topic_inferred_from_citation_fragment	Pearl mediation analysis (n.d.) - Resolved as a foundational topic rather than a single metadata record: causal mediation and intervention calculus. Use as conceptual background and repair the exact bibliographic target before final publication.	R0050
S2e5ed8d640	unresolved_external_reference / topic_inferred_from_citation_fragment	Per the program's framing literature stack (cited tight, 12-16 per paper) (n.d.) - No reliable external metadata/abstract was recovered automatically. Keep this row in the running bibliography-repair queue.	R0082
S6b1541e21d	unresolved_external_reference / topic_inferred_from_citation_fragment	Phenomenology and the Cognitive Sciences, 4(4), 429-452 (n.d.) - No reliable external metadata/abstract was recovered automatically. Keep this row in the running bibliography-repair queue.	R0059
S6d7f7343cf	manual_foundational_reference / topic_inferred_from_citation_fragment	Predictive Processing Model of Perception and Action for Self-Other Distinction Homeostatic active inference + enactive adaptivity: (n.d.) - Resolved as a foundational topic rather than a single metadata record: active inference and precision/value-of-information framing. Use as conceptual background and repair the exact bibliographic target before final publication.	R0093
S41f4bff29f	manual_foundational_reference / topic_inferred_from_citation_fragment	Problems of monetary management: the U (1975) - Resolved as a foundational topic rather than a single metadata record: measurement collapse under optimization pressure. Use as conceptual background and repair the exact bibliographic target before final publication.	R0059
Saf85a7ace6	manual_foundational_reference / topic_inferred_from_citation_fragment	Quantifying Epistemic Uncertainty in Deep Learning BALD / information gain / active inference epistemic value: (n.d.) - Resolved as a foundational topic rather than a single metadata record: active inference and precision/value-of-information framing. Use as conceptual background and repair the exact bibliographic target before final publication.	R0086
S6ae0b0cf55	resolved_with_abstract / abstract_read	ReAct: Synergizing Reasoning and Acting in Language Models (2022) - While large language models (LLMs) have demonstrated impressive capabilities across tasks in language understanding and interactive decision making, their abilities for reasoning (e.g. We apply our approach, named ReAct, to a diverse set of language and decision making tasks and demonstrate its effectiveness over state-of-the-art baselines, as well as improved human interpretability and trustworthiness over methods without reasoning or acting components.	http://arxiv.org/abs/2210.03629v3
S1de961dd50	resolved_with_abstract / abstract_read	Reflexion: Language Agents with Verbal Reinforcement Learning (2023) - We propose Reflexion, a novel framework to reinforce language agents not by updating weights, but instead through linguistic feedback.	http://arxiv.org/abs/2303.11366v4
S69b731796f	manual_foundational_reference / topic_inferred_from_citation_fragment	Refractory periods in neural firing: temporary suppression after activation (n.d.) - Resolved as a foundational topic rather than a single metadata record: post-activation suppression in neural systems. Use as conceptual background and repair the exact bibliographic target before final publication.	R0041
S52e67c0f31	manual_foundational_reference / topic_inferred_from_citation_fragment	Refractory periods in spiking neurons: temporary action threshold elevation post-firing (n.d.) - Resolved as a foundational topic rather than a single metadata record: post-activation suppression in neural systems. Use as conceptual background and repair the exact bibliographic target before final publication.	R0031
S81724e2b04	resolved_with_abstract / abstract_read	Representation Learning of Geometric Trees (2024) - To address this, we introduce a new representation learning framework tailored for geometric trees.	http://arxiv.org/abs/2408.08799v1
S1a878a46cb	resolved_with_abstract / abstract_read	Neural Language of Thought Models (2024) - In this work, we introduce the Neural Language of Thought Model (NLoTM), a novel approach for unsupervised learning of LoTH-inspired representation and generation.	http://arxiv.org/abs/2402.01203v2

ID	Status	Title / evidence note	Source / URL
S276b43e362	resolved_with_abstract / abstract_read	Why LLMs Fail at Causal Discovery and How Interventional Agents Escape (2026) - Recent benchmarks show that even fine-tuned models plateau on simple causal graphs and degrade as complexity grows, but why they fail has not been established. We propose Agentic Causal Bayesian Optimization (A-CBO), wherein a frozen language model serves as an interventional oracle answering targeted queries about intervention effects, while an external Bayesian loop concentrates beliefs over candidate graphs in logarithmically many rounds.	http://arxiv.org/abs/2605.27567v1
S1508b0c1de	manual_foundational_reference / topic_inferred_from_citation_fragment	Sein und Zeit (Being and Time) (1927) - Resolved as a foundational topic rather than a single metadata record: being-in-the-world and practical involvement. Use as conceptual background and repair the exact bibliographic target before final publication.	R0059
S512ba02392	unresolved_external_reference / topic_inferred_from_citation_fragment	Sense of agency under environmental drift: literature on agency maintenance through changing sensorimotor contingencies (n.d.) - No reliable external metadata/abstract was recovered automatically. Keep this row in the running bibliography-repair queue.	R0076
S312f10b4da	unresolved_external_reference / topic_inferred_from_citation_fragment	Sense of agency: Comparator model literature; Predictive Processing Model for Self-Other Distinction (n.d.) - No reliable external metadata/abstract was recovered automatically. Keep this row in the running bibliography-repair queue.	R0108
Sd1e7d7da2d	unresolved_external_reference / topic_inferred_from_citation_fragment	Sharp Minima Can Generalize for Deep Nets (2017) - No reliable external metadata/abstract was recovered automatically. Keep this row in the running bibliography-repair queue.	R0119
Sb418f35a5e	resolved_with_abstract / abstract_read	Simple and Scalable Predictive Uncertainty Estimation using Deep Ensembles (2016) - We propose an alternative to Bayesian NNs that is simple to implement, readily parallelizable, requires very little hyperparameter tuning, and yields high quality predictive uncertainty estimates. Through a series of experiments on classification and regression benchmarks, we demonstrate that our method produces well-calibrated uncertainty estimates which are as good or better than approximate Bayesian NNs.	http://arxiv.org/abs/1612.01474v3
S7e6f3322fc	manual_foundational_reference / topic_inferred_from_citation_fragment	Simulating homeostatic, allostatic and goal-directed forms of interoceptive control using Active Inference (n.d.) - Resolved as a foundational topic rather than a single metadata record: active inference and precision/value-of-information framing. Use as conceptual background and repair the exact bibliographic target before final publication.	R0094
Scc19e5f088	resolved_with_abstract / abstract_read	Structuring Open-Ended NAS: Semi-Automated Design Knowledge Structuring with LLMs for Efficient Neural Architecture Search (2026) - To address these issues, we propose to semi-automatically structure model design knowledge to guide the search process. To efficiently explore this vast space, we introduce FairNAD, using a multi-type mutation that enables broad exploration through mutation with fair idea sampling, Pareto-aware mutation, LLM-driven iterative mutation, and a fine-grained feedback loop.	http://arxiv.org/abs/2605.19247v1
S83d3d6303f	resolved_with_abstract / abstract_read	SWE-bench: Can Language Models Resolve Real-World GitHub Issues? (2023) - We find real-world software engineering to be a rich, sustainable, and challenging testbed for evaluating the next generation of language models. To this end, we introduce SWE-bench, an evaluation framework consisting of \$2,294\$ software engineering problems drawn from real GitHub issues and corresponding pull requests across \$12\$ popular Python repositories.	http://arxiv.org/abs/2310.06770v3
S8160897679	unresolved_external_reference / topic_inferred_from_citation_fragment	Synthese, 201(75). (n.d.) - No reliable external metadata/abstract was recovered automatically. Keep this row in the running bibliography-repair queue.	R0059
S3901111826	resolved_with_abstract / abstract_read	Technological Approach to Mind Everywhere: An Experimentally-Grounded Framework for Understanding Diverse Bodies and Minds (2021) - In this Perspective, I introduce TAME: Technological Approach to Mind Everywhere- a framework for understanding and manipulating cognition in unconventional substrates.	https://www.semanticscholar.org/paper/253c705ac587cb025a70c3b1093afe1f5126bcad
S6c700ea551	unresolved_external_reference / topic_inferred_from_citation_fragment	The contrast loss leverages paired observations under high-hazard vs low-hazard history to directly supervise the mediated component, with anti-cheat controls (shuffled pairs, wrong-history contrast) ensuring the result is not generic regularization (n.d.) - No reliable external metadata/abstract was recovered automatically. Keep this row in the running bibliography-repair queue.	R0058
Sb1120a0860	manual_foundational_reference / topic_inferred_from_citation_fragment	The Ecological Approach to Visual Perception (1979) - Resolved as a foundational topic rather than a single metadata record: ecological perception and affordances. Use as conceptual background and repair the exact bibliographic target before final publication.	R0059
S1f099d8a22	manual_foundational_reference / topic_inferred_from_citation_fragment	The free-energy principle: a unified brain theory? Nature Reviews Neuroscience, 11(2), 127-138 (2010) - Resolved as a foundational topic rather than a single metadata record: free-energy principle and active inference. Use as conceptual background and repair the exact bibliographic target before final publication.	R0059
Sd473132d35	unresolved_external_reference / topic_inferred_from_citation_fragment	The Metric Stack of Concern: From Viability Prediction to Maintained Self/World Boundaries in Minimal Agents (2026) - No reliable external metadata/abstract was recovered automatically. Keep this row in the running bibliography-repair queue.	R0015;R0111
S387c8c514c	unresolved_external_reference / topic_inferred_from_citation_fragment	The Minimum Description Length Principle (2007) - No reliable external metadata/abstract was recovered automatically. Keep this row in the running bibliography-repair queue.	R0135
Sac5208e823	manual_foundational_reference / topic_inferred_from_citation_fragment	The Phenomenon of Life: Toward a Philosophical Biology (1966) - Resolved as a foundational topic rather than a single metadata record: organism-centered value and biological normativity. Use as conceptual background and repair the exact bibliographic target before final publication.	R0059

ID	Status	Title / evidence note	Source / URL
S39f729145e	resolved_with_abstract / abstract_read	Toolformer: Language Models Can Teach Themselves to Use Tools (2023) - In this paper, we show that LMs can teach themselves to use external tools via simple APIs and achieve the best of both worlds. We introduce Toolformer, a model trained to decide which APIs to call, when to call them, what arguments to pass, and how to best incorporate the results into future token prediction.	http://arxiv.org/abs/2302.04761v1
S69e3aecee8	resolved_with_abstract / abstract_read	Toward Causal Representation Learning (2021) - Finally, we delineate some implications of causality for machine learning and propose key research areas at the intersection of both communities.	https://openalex.org/W3135588948
S5f79941c2c	resolved_with_abstract / abstract_read	Tree of Thoughts: Deliberate Problem Solving with Large Language Models (2023) - To surmount these challenges, we introduce a new framework for language model inference, Tree of Thoughts (ToT), which generalizes over the popular Chain of Thought approach to prompting language models, and enables exploration over coherent units of text (thoughts) that serve as intermediate steps toward problem solving. Our experiments show that ToT significantly enhances language models' problem-solving abilities on three novel tasks requiring non-trivial planning or search: Game of 24, Creative Writing, and...	http://arxiv.org/abs/2305.10601v2
S9040a88e8a	unresolved_external_reference / topic_inferred_from_citation_fragment	Underspecification Presents Challenges for Credibility in Modern Machine Learning (2022) - No reliable external metadata/abstract was recovered automatically. Keep this row in the running bibliography-repair queue.	R0118;R0160
S24a406bc4c	unresolved_external_reference / topic_inferred_from_citation_fragment	Universal Artificial Intelligence: Sequential Decisions Based on Algorithmic Probability (2005) - No reliable external metadata/abstract was recovered automatically. Keep this row in the running bibliography-repair queue.	R0122
Sa06325970b	manual_foundational_reference / topic_inferred_from_citation_fragment	Value of information in active inference: explicit treatment of intervention value vs information value Honest framing line: > Paper 23A redefines the program's probe-value signal as marginal MAE reducibility - the expected benefit of one more identifying intervention - and introduces a re-engagement mechanism that detects boundary staleness via non-null prediction surprise (n.d.) - Resolved as a foundational topic rather than a single metadata record: active inference and precision/value-of-information framing. Use as conceptual background and repair the exact bibliographic target before final publication.	R0077
Sf7d52174e7	unresolved_external_reference / topic_inferred_from_citation_fragment	Variational information maximisation for intrinsically motivated reinforcement learning (2015) - No reliable external metadata/abstract was recovered automatically. Keep this row in the running bibliography-repair queue.	R0059
Se3bc961bd1	manual_foundational_reference / topic_inferred_from_citation_fragment	Vervaeke relevance realization (lecture series) - relevance under representational bounds Plus the philosophical correlates the program operationalizes: Heidegger, Gibson, Uexkll, enactivism (Maturana/Varela/Thompson/Di Paolo), cybernetics (Ashby), active inference (Friston), pragmatism (Dewey), Canguilhem, Jonas, Simondon (n.d.) - Resolved as a foundational topic rather than a single metadata record: active inference and precision/value-of-information framing. Use as conceptual background and repair the exact bibliographic target before final publication.	R0081
Sf56633987d	manual_foundational_reference / topic_inferred_from_citation_fragment	Vervaeke, Awakening from the Meaning Crisis (n.d.) - Resolved as a foundational topic rather than a single metadata record: relevance realization and meaning-making. Use as conceptual background and repair the exact bibliographic target before final publication.	R0140
S5683a1d8ec	manual_foundational_reference / topic_inferred_from_citation_fragment	Viability Theory (1991) - Resolved as a foundational topic rather than a single metadata record: viability theory and constrained dynamical systems. Use as conceptual background and repair the exact bibliographic target before final publication.	R0111
Sc5c7f7e36c	resolved_with_abstract / abstract_read	Voyager: An Open-Ended Embodied Agent with Large Language Models (2023) - We introduce Voyager, the first LLM-powered embodied lifelong learning agent in Minecraft that continuously explores the world, acquires diverse skills, and makes novel discoveries without human intervention. Empirically, Voyager shows strong in-context lifelong learning capability and exhibits exceptional proficiency in playing Minecraft.	http://arxiv.org/abs/2305.16291v2
S7e990eba1d	resolved_with_abstract / abstract_read	Weakly supervised causal representation learning (2022) - We then introduce implicit latent causal models, variational autoencoders that represent causal variables and causal structure without having to optimize an explicit discrete graph structure. On simple image data, including a novel dataset of simulated robotic manipulation, we demonstrate that such models can reliably identify the causal structure and disentangle causal variables.	http://arxiv.org/abs/2203.16437v3
S9fe48dd9d0	unresolved_external_reference / topic_inferred_from_citation_fragment	weakly-supervised causal representation learning: interventional pairs unlock identifiability (n.d.) - No reliable external metadata/abstract was recovered automatically. Keep this row in the running bibliography-repair queue.	R0056
Sf5771e14e5	resolved_with_abstract / abstract_read	WebArena: A Realistic Web Environment for Building Autonomous Agents (2023) - The results demonstrate that solving complex tasks is challenging: our best GPT-4-based agent only achieves an end-to-end task success rate of 14.41%, significantly lower than the human performance of 78.24%.	http://arxiv.org/abs/2307.13854v4
Sb2ad4a427a	resolved_with_abstract / abstract_read	What Uncertainties Do We Need in Bayesian Deep Learning for Computer Vision? (2017) - There are two major types of uncertainty one can model. Aleatoric uncertainty captures noise inherent in the observations.	http://arxiv.org/abs/1703.04977v2
S6fe77e4d77	unresolved_external_reference / topic_inferred_from_citation_fragment	When Active Learning Fails, Uncalibrated Out-of-Distribution Uncertainty Epistemic uncertainty calibration: (n.d.) - No reliable external metadata/abstract was recovered automatically. Keep this row in the running bibliography-repair queue.	R0084
S23adffd29b	unresolved_external_reference / topic_inferred_from_citation_fragment	WILDS: A Benchmark of in-the-Wild Distribution Shifts (n.d.) - No reliable external metadata/abstract was recovered automatically. Keep this row in the running bibliography-repair queue.	R0165

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Sd9942e3f7e	resolved_with_abstract / abstract_read	World Models (2018) - We explore building generative neural network models of popular reinforcement learning environments. Our world model can be trained quickly in an unsupervised manner to learn a compressed spatial and temporal representation of the environment.	http://arxiv.org/abs/1803.10122v4